

Aditya Chunduri

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SUMMARY

Machine Learning Engineer with experience building and deploying computer vision and LLM-based systems. Strong background in PyTorch, Docker, and end-to-end ML pipelines spanning data acquisition, model training, evaluation, and inference optimization.

TECHNICAL SKILLS

Core ML: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, RAG
Languages: Python, C++, SQL, JavaScript, Bash
Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face
MLOps & Systems: Docker, Git, CI/CD, GitHub Actions, Flask, AWS
Data & Tools: PostgreSQL, ChromaDB, Neo4j, OpenCV, Pandas, NumPy

EXPERIENCE

Research Assistant Jun 2023 – Dec 2023
University of Southern California Los Angeles, California

- Architected artery-vein classification pipelines using U-Net, attaining a 0.92 Dice score on medical benchmarks
- Streamlined RMHAS image scraping and quality pipelines, reducing manual dataset curation time by 40%
- Programmed modular PyTorch evaluation pipelines to accelerate experimentation on domain-specific data by 30%

Research Intern Jun 2021 – Jul 2021
SSN College of Engineering Chennai, Tamil Nadu

- Conceptualized scalable Computer Vision pipelines for real-time tracking, processing 45 FPS and streamlining dataset generation
- Fine-tuned YOLO-based CNN models to increase detection robustness by 15% mAP across varied environmental conditions
- Coded reusable deep learning components in PyTorch and TensorFlow to standardize model development workflows

PROJECTS

AI Health Journal Jan 2024 – Present
Privacy-First Mental Health Assistant Python, Flask, Ollama, LangChain, ChromaDB

- Formulated a local RAG system using Ollama, enabling fully on-device inference with zero external API calls
- Executed DPO fine-tuning via a 76-prompt evaluation suite to align outputs with mental health safety guidelines
- Optimized ChromaDB indexing to ensure low-latency retrieval and real-time context injection

Akashic Tree Sep 2023 – Dec 2023
LLM Evaluation & Automated Content Pipeline Python, Ollama, Plotly, CI/CD

- Constructed CI/CD-driven benchmarking pipelines to evaluate local LLMs on reasoning and safety metrics
- Generated interactive dashboards utilizing Plotly to visualize and seamlessly compare model performance trends

MetaLearnML May 2023 – Aug 2023
Dataset-Agnostic AutoML Engine Python, PyTorch, Scikit-learn, FastAPI, Neo4j

- Constructed dataset-agnostic pipelines to infer task types and generate mechanized preprocessing workflows
- Deployed a meta-learning ranker to optimize model selection across 16 PyTorch and Scikit-learn configurations
- Tracked experiments using Neo4j and exposed performance metrics via a custom FastAPI interface

EDUCATION

University of Southern California (USC) Los Angeles, California
Master of Science (MS) in Applied Data Science Jan 2024 – Dec 2025

- *Coursework:* Machine Learning, Information Retrieval, Data Mining, Algorithms, Data Management, Data Engineering

SRM Institute of Science and Technology Chennai, Tamil Nadu
B.Tech in Computer Science and Engineering Jun 2019 – Jul 2023

- *Specialization:* Artificial Intelligence and Machine Learning

PUBLICATIONS

Wind Power Analysis Using Machine Learning in Wind Turbines Aug 2023
International Journal for Research in Applied Science & Engineering Technology (IJRASET)

- Proposed a neural network framework integrating numerical weather data to predict wind energy generation